

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
4	(a)			sodium zinc iron lead			1	1		1
	(b)	(i)		award (2) for all points plotted correctly tolerance $\pm \frac{1}{2}$ square award (1) for any 4 correct (1) suitable straight line drawn (with ruler) (1)		3		3	3	
		(ii)		award (1) for either of following - order of reactivity is $\text{Mg} > \text{Al} > \text{Zn} > \text{Cu}$ - magnesium and aluminium are more reactive than zinc and copper is less reactive than zinc – must refer to all four metals award (1) for any of following - copper does not react with zinc chloride / does not displace zinc - temperature doesn't change/increase with copper - aluminium and magnesium react with zinc chloride / displace zinc award (1) for any of following - reaction between magnesium and zinc chloride is more exothermic than that between aluminium and zinc chloride - magnesium reaction more exothermic than aluminium - temperature increases more with magnesium than aluminium - magnesium most exothermic			3	3		3
				Question 4 total	0	3	4	7	3	4